BMXPARTSDEPOT.COM · PRODUCTION SPEC

FITMENT GUIDE DIAGRAM SPECS

13 diagrams
1 delivered · 12 open
2026-08-28

Every diagram placeholder across the ten guides, with what it has to show and the canvas to draw it on. **One rule runs through all of them:** a diagram may show where a measurement is taken, but it may not state the measurement unless that figure survived the source-verification pass.

RULES FOR ALL THIRTEEN

FOLDER	assets/diagrams/ in the repo
RENDERED AT	718px wide maximum, 238px minimum. It never renders wider than 718.
BACKGROUND	Sits on a white card. Design for light, do not rely on a dark ground.
TEXT	All outlined to paths. Webfonts do not load inside an <img>, so live text loses Bebas Neue.
TYPE FLOOR	Nothing below 30 viewBox units. Headings 44, secondary 32 to 38.
PALETTE	ink #08090b material · cyan #00a4e6 the moving or key part · orange #e64700 callouts and the thing to notice · #6b7280 dimension lines · #e2e5e9 hairlines. No new colors.
INSIDE THE FILE	A <title> and <desc>. Pure vector, no embedded raster.

WHY MOST OF THESE CARRY NO NUMBERS

Only the bottom bracket, spindle, axle and dropout-spacing figures were verified against manufacturer sources. Pedal threads, seat post diameters, bar clamp sizes, headset dimensions, fork offsets and era year ranges were **not**, and research could not source them. So those diagrams teach **where to measure and what the parts are**, and the numbers stay in the page tables where they carry their sourcing badges. A diagram with an invented dimension is worse than no diagram, because a drawing reads as with an invented dimension is worse than no diagram, because a drawing reads as authoritative in a way a sentence does not.

THE THIRTEEN

01 BOTTOM BRACKET SHELL COMPARISON

DELIVERED

`bmx-bottom-bracket-shell-comparison.svg`

Bottom Brackets and Spindles · #identify

Specced and drawn. Four shells to a 5-units-per-mm scale, bearing position shown, the 41.3mm bearing identical on American and Mid.

02 THE THREE REAR HUB TYPES IN SECTION

SPECCED

`bmx-rear-hub-types-comparison.svg`

Wheels, Hubs and Axles · #identify-rear-hub

Freewheel, cassette and freecoaster cut along the axle, same visual scale. Show what is attached to what: the freewheel body threads onto the shell and carries its own ratchet, the cassette keeps the ratchet in the hub with a splined driver, the freecoaster adds a clutch and a slack gap. **Draw the slack gap explicitly** with an arrow for the direction of free travel, in orange.

LABEL TEXT

- `Freewheel` / threads onto the hub shell
- `Cassette` / ratchet inside the hub, splined driver
- `Freecoaster` / clutch and slack, coasts backwards

Omit: every dimension. Driver spline counts are brand-specific and unverified, slack is adjustable and varies by model.

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viewBox `0 0 1000 480, three panels across, hubs ~280 units wide each`

03 BUTTED TUBE WALL THICKNESS IN SECTION

TO DRAW

`bmx-butted-tubing-section.svg`

Frame ID, Materials and Geometry · #butted-tubing

Two tubes cut lengthwise. A straight gauge tube with even wall the whole way, and a double butted tube with thick walls at each end thinning through the middle. Exaggerate the difference so it reads, then **label the drawing "not to scale"** in the corner, because wall thicknesses were never sourced and an exaggerated section that looks dimensioned is a lie.

LABEL TEXT

- `Straight gauge` / same wall throughout
- `Double butted` / thicker at the joints, thinner in the middle
- `not to scale` in the corner, small and grey

Omit: any wall thickness figure, any weight claim, any tubing brand.

viewBox `0 0 1000 420, two tubes stacked, cyan for the butted walls`

04 WHERE THE FOUR FRAME MEASUREMENTS ARE TAKEN

TO DRAW

bmx-frame-measurement-points.svg

Frame ID, Materials and Geometry · #top-tube-length

Side view of a bare BMX frame with four measurements marked as dimension lines with end caps. This is one of the most useful drawings on the site, because "measure your top tube" is meaningless until someone shows you the two points.

THE FOUR, EXACTLY

- `Top tube length` centre of head tube to centre of seat tube, horizontal
- `Chainstay length` bottom bracket centre to rear axle
- `Standover` ground to top of top tube
- `Wheelbase` front axle to rear axle

Omit: all figures and all ranges. Geometry numbers were not sourced. Ghost the wheels in hairline so the axles have somewhere to be without the drawing becoming a bike.

viewBox `0 0 1000 620, frame in ink, dimension lines in orange`

05 THE FOUR PARTS OF A BMX DRIVETRAIN

TO DRAW

`bmx-drivetrain-anatomy.svg`

Drivetrain, Chains and Sprockets · #overview

Two views. A side view naming sprocket, chain and driver, and a small top-down view showing the chainline as a straight plane running from sprocket to driver. The top view is the one that matters: chainline is impossible to explain from the side and obvious from above.

LABEL TEXT

- Side: `Sprocket`, `Chain`, `Driver`
- Top: `Chainline` with a dashed straight line, and `out of line` on a second faint sprocket offset from it

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Omit: tooth counts, gear ratios, chainline distance in mm. All unsourced.

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viewBox `0 0 1000 560, side view upper two thirds, top view lower third`

06 ONE, TWO AND THREE-PIECE CRANKS COMPARED

TO DRAW

`bmx-crank-types-comparison.svg`

Cranks and Pedals · #crank-types

Three cranks, same scale, drawn so the **number of separate pieces is countable at a glance**. That is the whole diagnostic. One-piece is a single S-shaped forging through the shell. Two-piece has the spindle fixed to the drive arm with the second arm bolting on. Three-piece has two arms and a separate spindle. Use colour to separate the pieces within each crank rather than to decorate.

LABEL TEXT

- `One-piece` / Ashtabula, one forging
- `Two-piece` / spindle fixed to the drive arm
- `Three-piece` / two arms, separate spindle
- A small `1` `2` `3` count marker on the pieces themselves

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Omit: arm lengths, spindle diameters, pedal thread sizes. Pedal threading is the biggest unverified item on that page.

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viewBox `0 0 1000 460, three across`

07 HOW A U-BRAKE PULLS

TO DRAW

`bmx-u-brake-action.svg`

Brakes, Cables and Detanglers · #u-brake

Front-on view of a U-brake with the two arms crossing over their posts, the straddle cable forming a triangle up to the yoke, and the main cable pulling from the yoke. Arrows for the direction of force at each point: up at the yoke, in along each straddle leg, inward at each pad. The crossover is the thing people get wrong when reassembling one.

LABEL TEXT

- `Yoke`, `Straddle cable`, `Arms cross over`, `Pad`, `Post`

Omit: post spacing, straddle height, cable lengths. All unverified and all adjustable.

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viewBox 0 0 1000 640, arms and cable in ink, force arrows in orange, rim ghosted in hairline

08 HEADSET TYPES IN CROSS SECTION

TO DRAW

`bmx-headset-types-section.svg`

Headsets, Forks and Stems · #bmx-headset-standards

Three head tubes cut vertically, same scale, with **the bearing drawn in its actual position in each**. Integrated sits directly in the head tube. Zero stack recesses into the head tube. Traditional pressed cup sits in a cup that protrudes above and below. The bearing position is the identification method, so it has to be the visual difference.

LABEL TEXT

- `Integrated` / bearing sits in the head tube
- `Zero stack` / bearing recessed into the head tube
- `Traditional` / bearing in a cup pressed into the head tube

Omit: every dimension, including the 45/45 bearing angle and any stack height. None were sourced.

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viewBox 0 0 1000 520, three across, bearings in cyan, cups in orange

09 THREADLESS COMPRESSION, EXPLODED

TO DRAW

`bmx-threadless-compression-exploded.svg`

Headsets, Forks and Stems · #threadless-headset

Vertical exploded view along the steerer, parts separated in assembly order with a dashed centreline through them. The point is that the top cap pulls the compression plug, which pulls the stack together, and the stem bolts then hold it. Show the pull with an arrow up the centreline from plug to cap.

ORDER, BOTTOM TO TOP

- `Lower bearing`, `Head tube`, `Upper bearing`, `Compression ring`, `Dust cover`, `Spacers`, `Stem`, `Top cap`
- `Compression plug` shown inside the steerer, ghosted through it

Omit: steerer diameter, stack heights, bolt sizes.

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viewBox `0 0 620 900, portrait. This is the one tall diagram, it will render narrow and that is correct.`

10 FORK OFFSET AND TRAIL

TO DRAW

`bmx-fork-offset-and-trail.svg`

Headsets, Forks and Stems · #fork-offset

Side view of a front end with the steering axis drawn as a dashed line extended all the way to the ground. Offset is the perpendicular distance from that axis to the axle. Trail is the distance along the ground from where the axis lands to the tyre contact patch. Two different measurements that get used interchangeably, which is exactly why the drawing is needed.

LABEL TEXT

- `Steering axis` on the dashed line
- `Offset` perpendicular, axis to axle
- `Trail` along the ground
- `Contact patch` at the tyre

Omit: offset figures, head angle degrees, wheel size. Unsourced, and the relationship is what matters.

viewBox `0 0 760 720, fork in ink, axis dashed grey, offset and trail in orange`

11 THE FOUR BAR MEASUREMENTS

TO DRAW

`bmx-handlebar-measurement-points.svg`

`Handlebars, Grips and Cockpit · #bar-numbers`

Bars from the front for rise and width, and a small top-down inset for backsweep and upsweep, since the two sweeps are invisible from the front. Sweeps drawn as angles off a reference line rather than as figures.

LABEL TEXT

- Front view: `Rise`, `Width`
- Top inset: `Backsweep`, `Upsweep`

Omit: all figures and ranges, and the clamp diameter. Also worth knowing while you draw: sources do not agree whether rise is measured to the tube or the grip centreline, so **show the measurement running to the top of the tube** and leave it at that.

viewBox `0 0 1000 600, front view main, top inset lower right`

12 THE THREE SEAT INTERFACES

TO DRAW

`bmx-seat-interfaces-comparison.svg`

`Seats, Posts and Clamps · #seat-types`

Three cross sections showing how the seat meets the post in each system. Pivotal: a splined boss on the post head with a single bolt down through the seat. Railed: two rails clamped between a two-part post head. Tripod: a three-point plate. The bolt count and direction is the fast field identification, so make it obvious.

LABEL TEXT

- `Pivotal` / one bolt down through the seat, splined angle adjustment
- `Railed` / two rails clamped in the post head
- `Tripod` / three-point mounting plate

Omit: post diameters, rail spacing, spline counts. Whether pivotal splines interchange between brands is an open question on that page, so do not draw anything implying a universal spline.

viewBox `0 0 1000 480, three across, post in ink, seat base in cyan, bolt in orange`

13 **ERA FEATURE SEQUENCE** TO DRAW

`bmx-era-feature-sequence.svg`

Era, Dating and Restoration ID · #era-boundaries

This one has a trap in its own title. The page's argument is that the old school, mid school and modern year boundaries are repeated everywhere and could not be traced to a primary source. So this cannot be a dated timeline. Draw it as horizontal bands, one per component group, showing standards succeeding each other left to right, with the transitions as **soft overlapping gradients rather than hard lines**. Three era names float above as wide, deliberately vague zones.

ROWS

- Bottom bracket: American → Euro → Mid → Spanish
- Headset: threaded → threadless → integrated
- Cranks: one-piece → three-piece → two-piece
- Brakes: caliper and U-brake → U-brake and gyro → often none